Assignment 5 - Exploring Buildings and Roads with Overture Maps

Submission Date: By Tuesday November 10th 11.59pm

This assignment uses the Week 9 notebook on downloading data and spatial processing. You are helping a local planning team describe buildings and roads in a small study area. Use the overturemaps Python package to download data, process it in GeoPandas, and answer a spatial question.

Complete the work in a new Google Colab notebook. Put your name, course number, and assignment title at the top. Label every part clearly, so run every code cell and leave all outputs visible for review. Use descriptive snake_case names and include units where appropriate.

Table 1 outlines the required assignment components.

Component	Requirements	Points
1. Choose an area and download data	Choose a small urban study rectangle (not Fairfax or GMU), roughly 1-4 km across, and state its location and why you chose it. Define bbox = (west, south, east, north) in longitude/latitude decimal degrees. Download building and segment data using geodataframe(), retaining segments where subtype is "road". Show the first five rows and feature count of each layer. Save separate GeoPackages, retaining id and geometry for buildings and id, class, and geometry for roads.	20
2. Prepare the study layers	Check each layer's CRS, geometry types, and missing, empty, or invalid geometries (if any). Remove any repeated IDs. Transform both layers and a box(*bbox) study polygon to a suitable local CRS in meters (EPSG:32618 is suitable near Fairfax). Based on the intersection approach from week 8, clip both layers to the study polygon. Print the remaining counts and study area in km2. Explain why to_crs() is needed before measuring.	20
3. Describe the buildings	Using the clipped polygons, calculate the building count and building count density. Then calculate the building footprint area and footprint density (footprint records per km^2). Show a table of the five largest clipped footprints with ID and area (or all if fewer than five remain). Map the buildings for your clipped area and show the study boundary, with the building footprint area on the color scale. Include a title and color legend with units. Why footprint area does not measure floor area or population.	20
4. Measure the road network	Calculate the total clipped road segment length in km for your bounding box area. Create a table of segment count and total length by class, including a "Not recorded" group for missing class values. Map road classes with the buildings and study boundary in the background, and include a title and legend. Report footway and cycleway lengths separately, reporting zero if either class is absent. In two sentences, explain why the segment count differs from the length, and why a missing cycleway class does not prove cycling is impossible.	20
5. Answer a planning question	Write one question before analyzing it, such as "What share of mapped road length is classified as footway?" or "How many mapped buildings have at least 2,000 m^2 inside the study area?" Use Python to calculate the answer and show a small result table plus a map highlighting the relevant features (or reporting that none match). Write 100-150 words answering the question with numbers and units. Define the selection and denominator, and discuss data completeness and the rectangular boundary. Do not claim a complete inventory.	20

Table 1 Assignment components and requirements.

Submit the completed .ipynb file online. Restart the runtime and run all cells before submitting. Check that all five parts, tables, maps, and written answers are present and no unexpected errors remain. Calculate results in Python rather than typing them manually. Save and reload your extracts to avoid repeated downloads.

Also submit a .PDF file of your notebook (or a .PDF of screenshots), for easy review.

Table 2 outlines the assessment criteria. Component points total 100, with the criteria below describing how the work is assessed across those components.

Criterion	Evidence	Weight
Correctness / completeness	All five parts are present. Downloads, quality checks, clipped measurements, grouped road lengths, and the final answer are correct.	40%
Spatial data selection / use	Suitable data, geometry types, and a local projected CRS are used. Clipping, area, length, density, and selection rules match the stated study area.	25%
Documentation / readability	Names preferably follow snake_case. Code is readable, with maps and tables having clear labels and units. Written answers explain results and limitations.	15%
Testing / output	Visible outputs support the answers. Empty selections and missing classes are handled. All cells run in order and totals agree with the grouped summaries.	15%
Organization / submission	The notebook is clearly labeled, logically ordered, complete, and submitted as .ipynb and .PDF files.	5%

Table 2 Assessment criteria.

On this particular assignment, you are not allowed to use AI, otherwise it defeats the point of trying to learn this content. Use the Week 9 notebook, earlier GeoPandas examples, and official Python, GeoPandas, and Overture documentation for help. The Mason Honor Code applies.